University of Cape Town

**University Examinations – May/June 2024**
**Department of Mathematics and Applied Mathematics**

MAM3000W

**SOLUTIONS Module 3MS: Metric Spaces**

**Time : 2 hours** **Full Marks: 40**

1. (a) We repeatedly use the fact that $d$ is a metric on $X$. For $x, y, z \in X$:
(M1) $p(x, x) = d(f(x), f(x)) = 0$, and if $p(x, y) = 0$, then $d(f(x), f(y)) = 0$, so $f(x) = f(y)$; so $x = y$. It is in this last step that $f$ being injective is used.
(M2) $p(x, y) = d(f(x), f(y)) = d(f(y), f(x)) = p(y, x)$.
(M3) $p(x, z) = d(f(x), f(z)) \leq d(f(x), f(y)) + d(f(y), f(z)) = p(x, y) + p(y, z)$.

(b) $B(0, 4) = \{x \in \mathbb{R} : |e^x - e^0| < 4\} = (-\infty, \ln 5)$. [8+2=10]

2. ($\Longrightarrow$) If $\mathrm{bd}(A) = \emptyset$, then $A \cap \mathrm{bd}(A) = A \cap \emptyset = \emptyset$, so $A$ is open,
and again if $\mathrm{bd}(A) = \emptyset$, then $\overline{A} = A \cup \mathrm{bd}(A) = A \cup \emptyset = A$, so $A$ is closed.
($\Longleftarrow$) If $A$ is closed, $\overline{A} = A$ and also if $A$ is open, then $A'$ is closed and so $\overline{A'} = A'$,
thus $\mathrm{bd}(A) = \overline{A} \cap \overline{A'} = A \cap A' = \emptyset$. [6]

3. (a) True.
Since $A \subseteq A \cup B$, we have $\overline{A} \subseteq \overline{A \cup B}$, and similarly $\overline{B} \subseteq \overline{A \cup B}$;
so $\overline{A} \cup \overline{B} \subseteq \overline{A \cup B}$. For the other inclusion, Let $x \in \overline{A \cup B}$, then there exists a sequence $(x_n)$ in $A \cup B$ such that $x_n \to x$ as $n \to \infty$, we then have that either $A$ or $B$ contain infinitely many elements of sequence $(x_n)$, suppose without loss of generality that it is $A$, then we have that $A$ contains a subsequence $(x_{n_k})$ of $(x_n)$ and $x_{n_k} \to x$ as $n \to \infty$ and so $x \in \overline{A} \subseteq \overline{A} \cup \overline{B}$.

(b) True. Let $(x_n)$ be a Cauchy sequence in finite subset A of metric space $(X, d)$. If $A$ has at most one element, the result is trivially true, otherwise,
take $\varepsilon = \min\{d(x, y) | x \neq y\}$, then there exists $N \in \mathbb{N}$ such that if $m \geq n \geq N$, $d(x_m, x_n) < \varepsilon$, we then have that $(x_n)$ is eventually constant and converges to an element in $A$, thus $A$ is complete.

(c) Let $T = \{0, 1\}$. ($\Longrightarrow$) Suppose that $f : X \to T$ is continuous, but not constant. Then $f^{-1}(\{0\})$ is a clopen subset of $X$ which is neither empty nor $X$ itself. So $X$ is disconnected. ($\Longleftarrow$) Suppose $X$ is disconnected. Then there exists a clopen subset $A$ of $X$ such that $A \neq \emptyset$ and $A \neq X$. The characteristic function on $A$ is then a continuous function from $X$ to $T$, that is not constant.

[4+4+4=12]

4. Read this result, and answer the questions that follow:

**Proposition** If the metric space $(X, d)$ is compact, then this condition holds:
For any decreasing sequence $(A_n)$ of non-empty subsets of $X$,

$$\bigcap_{n\in\mathbb{N}} \overline{A}_n \neq \emptyset.$$

**Proof** Suppose $(A_n)$ is a decreasing sequence of non-empty subsets of $X$. (1)
For each $n \in \mathbb{N}$, take $a_n \in A_n$. (2)
There exists a subsequence $(a_{n_k})$ of $(a_n)$ such that $a_{n_k} \to x$ for some $x \in X$. (3)
We show that $x \in \overline{\bigcap_{n\in\mathbb{N}} A_n}$. (4)
Let $m \in \mathbb{N}$; we show $x \in \overline{A_m}$. (5)
Suppose $r > 0$. (6)
Take $N \in \mathbb{N}$ such that $k \geq N$ implies that $a_{n_k} \in B(x; r)$. (7)
Take $K \in \mathbb{N}$ such that $K \geq N$ and $n_K > m$. (8)
Then $B(x; r) \cap A_m \neq \emptyset$. (9)

(a) A sequence $(A_n)$ such that $A_1 \supseteq A_2 \supseteq \ldots \supseteq A_n \supseteq \ldots$,
that is, $A_m \supseteq A_n$ whenever $m < n$.

(b) This uses compactness of $(X, d)$.

(c) This uses $a_{n_k} \to x$.

(d) Since $n_k \geq k$, by the definition of a subsequence, it suffices to choose $K \geq N$ and $K \geq m$.

(e) We have $a_{n_K} \in A_{n_K} \subseteq A_m$, from line (8) and the fact that the sequence of subsets is decreasing.
Also $a_{n_K} \in B(x; r)$, from line (7).

[2+2+2+2+2=10]

5. ($\Longrightarrow$) Suppose $B(x,r)$ is an open ball in an ultrametric space. Then, for any $y,z \in B(x,r)$, we have $d(y,z) \leq \max\{d(y,x), d(x,z)\} < r$. Since $\mathrm{diam}B(x,r) = \sup\{d(y,z) : y,z \in B(x,r)\}$, we obtain $\mathrm{diam}B(x,r) \leq r$.

($\Longleftarrow$) Let $x,y,z \in X$. If any two or more of these elements are equal, the condition $d(x,z) \leq \max\{d(x,y), d(y,z)\}$ is trivially satisfied. So suppose $x,y,z$ are distinct. For each $n \in \mathbb{N}$, let $k_n = \max\{d(x,y), d(y,z)\} + \frac{1}{n}$. Then $x,z \in B(y,k_n)$ for all $n$. So, for each $n$,

$$d(x,z) \leq \mathrm{diam}B(y,k_n) \leq k_n.$$

Letting $n \to \infty$ gives $d(x,z) \leq \max\{d(x,y), d(y,z)\}$, as required.

[4]